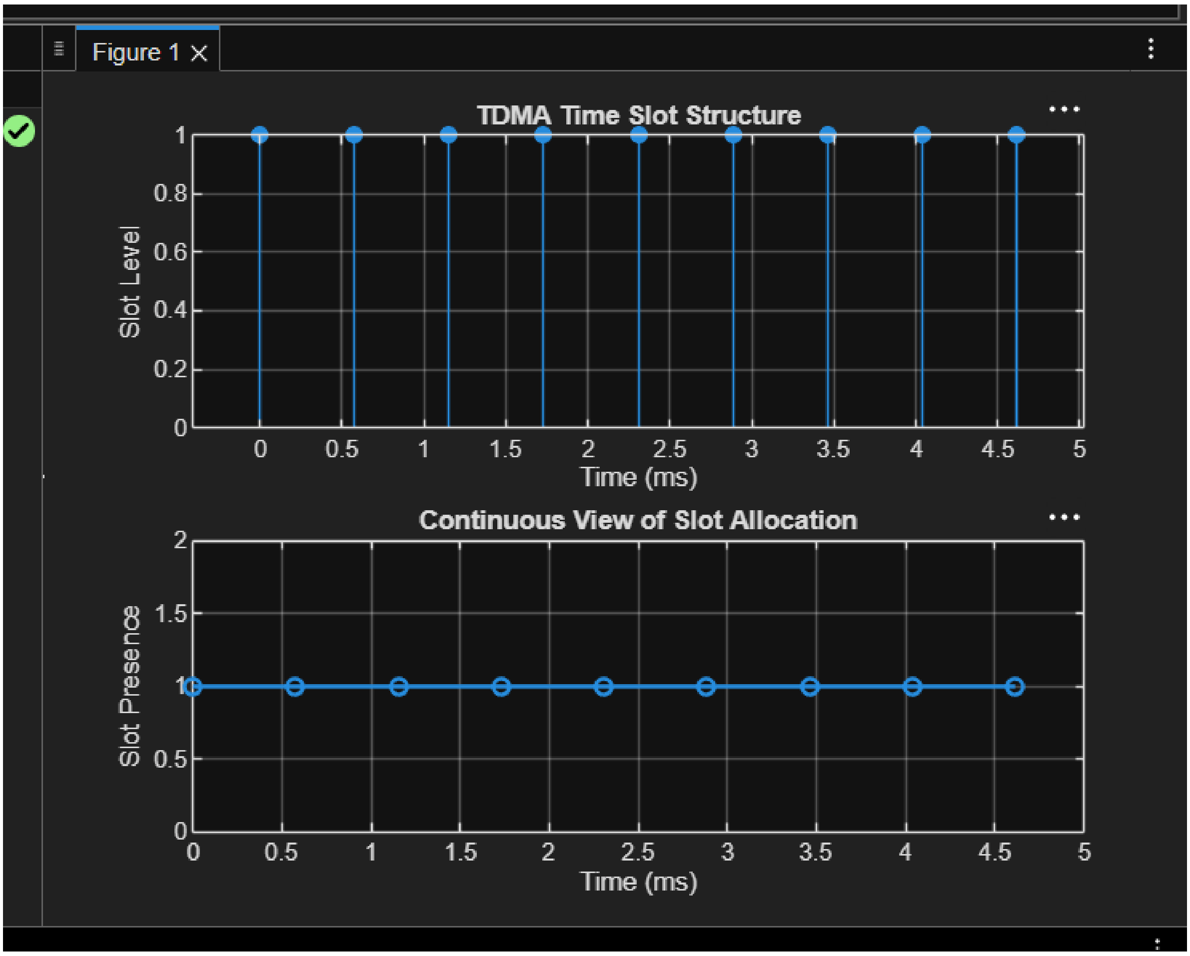
Ex.No. 2: Implementing TDMA Slot Allocation in GSM. Date

OBJECTIVE1:

CODE:

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;

t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```

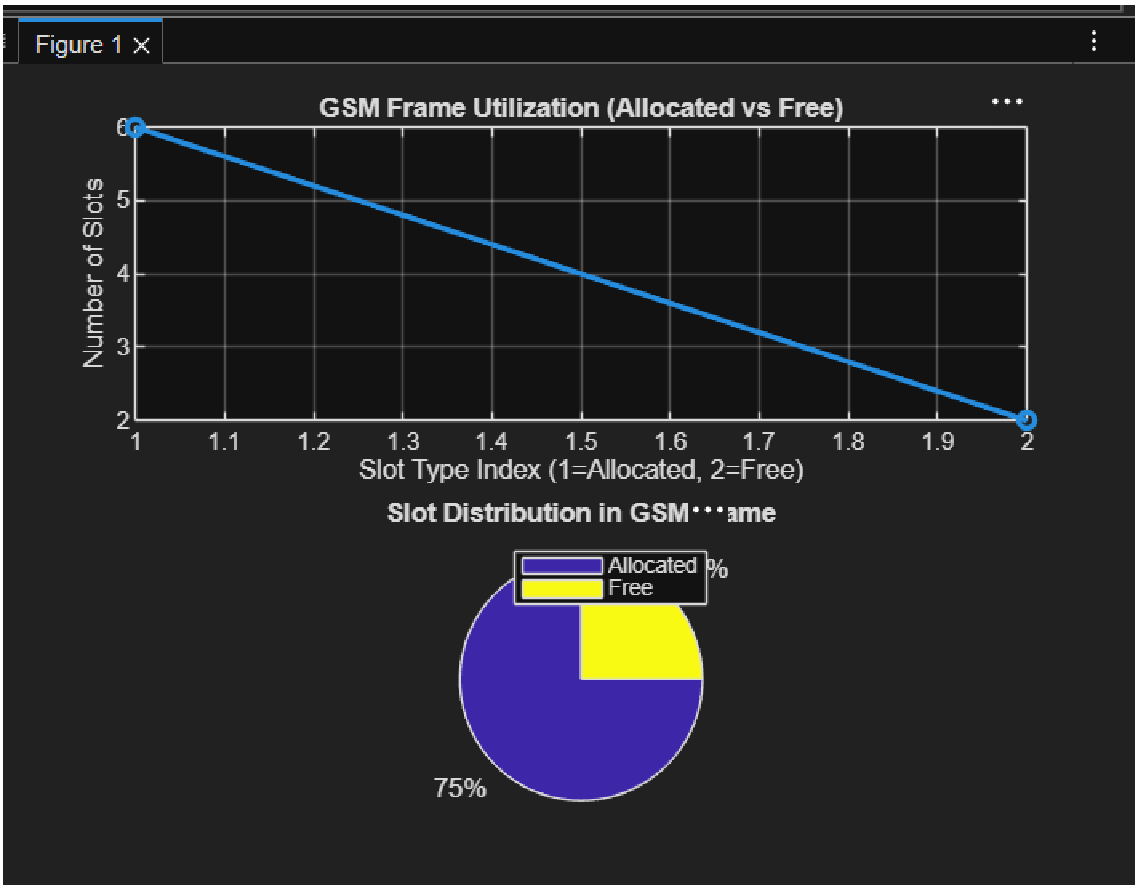


Objective 2:

Code:

```
clc; clear; close all;
TotalSlots = 8;
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;

Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
```

```
% Graph 1: Simple Line Plot (SAFE - no bar function)

subplot(2,1,1)

plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)

title('GSM Frame Utilization (Allocated vs Free)')

xlabel('Slot Type Index (1=Allocated, 2=Free)')

ylabel('Number of Slots')

grid on

% Graph 2: Pie Chart (SAFE)

subplot(2,1,2)

pie([AllocatedSlots FreeSlots])

title('Slot Distribution in GSM Frame')

legend('Allocated','Free')
```



OBJECTIVE 3:

CODE:

```
clc;
clear;
close all;

Users = 1:8;        % Total users
TotalSlots = 5;     % Available slots

Alloc = zeros(1,8);   % Initialize allocation
Alloc(1:TotalSlots) = 1;   % Assign slots to first users
Blocked = 1 - Alloc;      % Remaining users blocked

disp('User Wise Slot Allocation')
disp(table(Users', Alloc', ...
   'VariableNames', {'User','Status_Allocated'}))

figure

% Graph 1: Allocation per user
subplot(2,1,1)
stem(Users, Alloc, 'filled', 'LineWidth', 2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
```

```
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on

% Graph 2: System view
subplot(2,1,2)
plot(Users, Alloc, '-o', 'LineWidth', 2)
hold on
plot(Users, Blocked, '-s', 'LineWidth', 2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
```

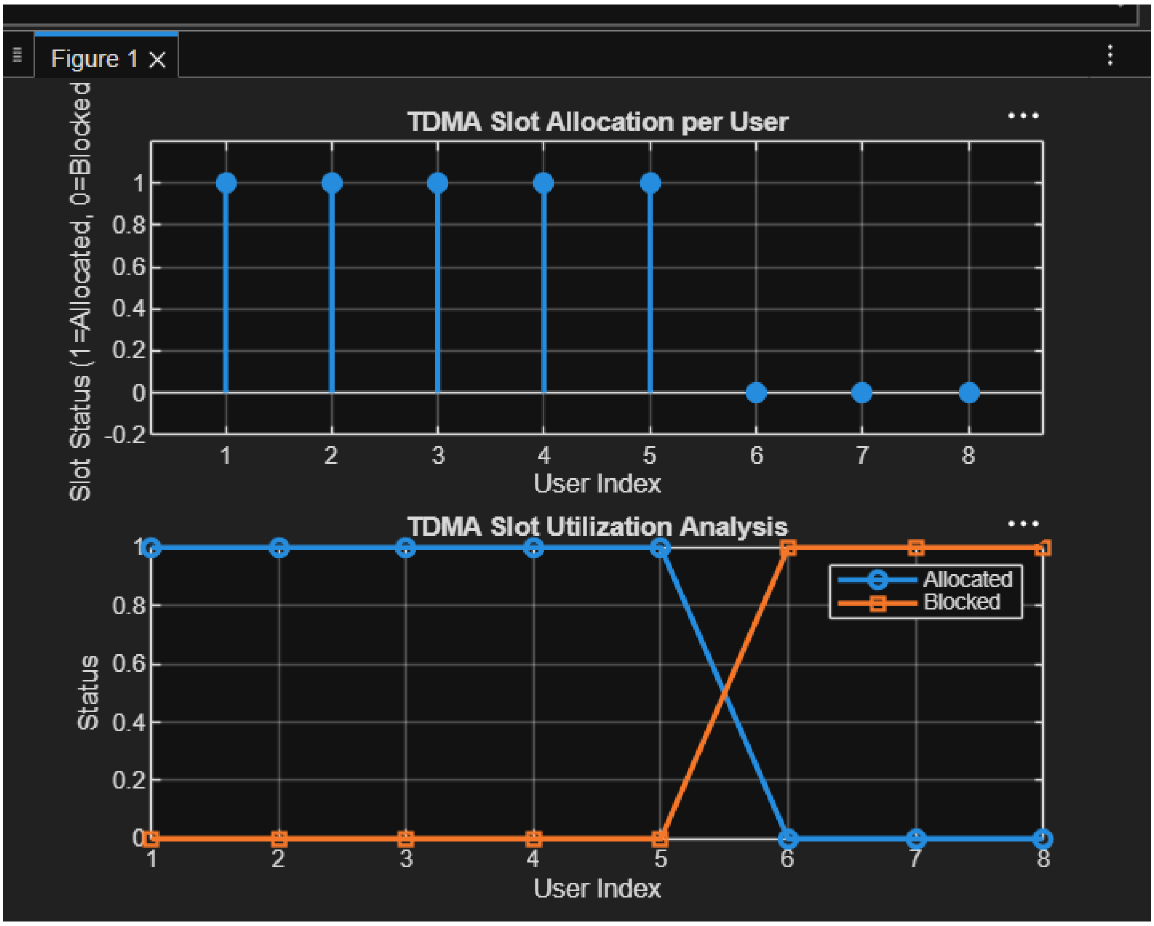


Figure 1
TDMA Slot Allocation per User
Slot Status (1=Allocated, 0=Blocked
1
0.8
0.6
0.4
0.2
0
-0.2
1 2 3 4 5 6 7 8
User Index
TDMA Slot Utilization Analysis
Status
1
0.8
0.6
0.4
0.2
0
Allocated
Blocked
1 2 3 4 5 6 7 8
User Index